Lecture 17: A Person in the World of People: Self and Other, Part II; Some Mysteries: Sleep, Dreams, and Laughter (full transcript)

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Overview: This lecture begins with the second half of the discussion on social psychology. Students will learn about several important factors influencing how we form impressions of others, including our ability to form rapid impressions about people. This discussion focuses heavily upon stereotypes, including a discussion of their utility, reliability, and the negative effects that even implicit stereotypes can incur. The second half of the lecture introduces students to two prominent mysteries in the field of psychology. First, students will learn what is known and unknown about sleep, including why we sleep, the different types of sleep, disorders, and of course, dreams, what they are about and why we have them. Second, this half reviews how laughter remains a mysterious and interesting psychological phenomenon. Students will hear theories that attempt to explain what causes us to laugh and why, with a particular emphasis on current evolutionary theory.

Chapter 1. Introduction [00:00:00]

Professor Paul Bloom: Just to review, here's where we left off. The discussion from last lecture and for about half of this lecture is going to be social psychology. And so, we started off by talking about certain fundamental biases in how we see ourselves. We then turned to talk about a bias and how we see other people, the fundamental attribution error. And now we're talking a little bit about some aspects of how we see other people. So, we quickly talked about certain aspects of why we like other people including proximity, similarity, and attractiveness, and where we left off was a discussion of the Matthew effect, which is basically that good things tend to compound. If you're rich you'll get a better education, if you're smart people will like you more, if you're attractive and so on. Nobody bring up their papers at this point. They'll collect them at the end of class. What I want to talk to — [laughter] Okay, except for you. Just hand me it now. [laughter] I'm going to ask the teaching fellows to stop anybody from approaching that area.

Chapter 2. First and Fast: How We Form Impressions of Others [00:01:39]

I want to begin by talking about [laughter] impression formation, how we form impressions of others, and tell you a couple of interesting things about impression formation. The first one is, first impressions matter a lot. They matter a lot for different reasons. They might matter a lot because humans have, in general, a confirmation bias such that once you believe something other information is then encoded along the likes to support what you believe. So, the classic study here was done by Kelley where a guest speaker comes in and some of the students received a bio describing the speaker as very warm, the other as — do not bring your paper up if you're coming in late. Just — at the end of class, yeah. [laughter] Others got a bio saying — thanks, Erik — the speaker was rather cold and then it turned out later on [laughter] when they're asked for their impressions of the speaker people are very much biased by what they first assumed. If I'm described to you as a vivacious and creative person and

you see me and I'm all kind of bouncing around and everything, you could then confirm this as, "Look how vivacious and creative he is." If I'm described as somebody who drinks too much, you might think he's an alcoholic. If he's described as somebody who's insecure and nervous, you could interpret my activity as nervous twitches. Your first impression sets a framework from which you interpret everything else.

This was the theme of an excellent movie called Being There starring Peter Sellers. And the running joke of the movie "Being There" was that the main character, the character Chauncey Gardner, somehow through accident had the reputation for being a genius but while, in reality, he was actually mildly retarded. But he would go around and people would ask him his opinions on politics and he would say things like "Well, I like being in the garden." And because of his reputation as a genius people said, "Wow. That's very profound. I wonder what he means." And — or people would talk to him and he'd just stare at them and say — and people would say — would be intimidated by his bold and impetuous stare when actually he just totally didn't know anything. So, first impressions can shape subsequent impressions not just when dealing with people.

A little while ago there was a sniper, actually a pair of snipers killing people in Washington and the one thing everybody knew about it was there was a white van involved. It turned out there was no white van at all but in the first incident somebody saw a white van, this was reported in all the newspapers, then every other incident people started seeing the white van. So, they started looking for them and they started to attending — attend to them. So, first impressions matter hugely when dealing with people because it sets the stage for how we interpret everything else.

A second finding building on the first is that we form impressions very fast, very quickly, and this is a literature known as "thin slices." The idea is you don't have to see much of a person to get an impression of what they are. The first studies done on this were actually done on teachers, on university professors. So, university professors have teaching evaluations and you could use this as a rough and ready approximation of what students think of them. So, what you do then is — the question that these people were interested in, Rosenthal and Ambady, two social psychologists, were how long do you have to look at a professor to guess how popular a teacher he is? So, they showed these clips for a full class. Do you have to see them for a full class? Do you have to see them for two classes? Do you have to see them for a half hour? How long do you have to be around a person to see him, to estimate how good a lecturer that person is? And the answer is five seconds. So, after clips of five seconds people are pretty good at predicting what sort of evaluations that person will have.

Remember "The Big Five," how we evaluate people on "The Big Five?" Well, you have a roommate and your roommate you could evaluate on "The Big Five." You've had a lot of experience with him or her. How much time do you need to evaluate somebody on the five dimensions of personality? The answer is, again, not much time at all. After very brief exposures to people, people are very accurate at identifying them on "The Big Five." One of the more surprising findings is — concerns sexual orientation or "gaydar." That's not a scientific term [laughter] but the same psychologists were interested in studying how quickly you can — if at all how long does it take to figure out somebody's sexual orientation?

Now, what they did was — they were clever psychologists so they set it up in a study where the people did not know sexual orientation was at issue. So, for instance, they may be people like you who filled in

a form, one question along a very long form was your sexual orientation, and then you're sitting down being interviewed by somebody and your interview is being filmed, and then other people are shown — who don't know you are shown the film. And the finding is that people based on thin slices are quite good at detecting sexual orientation. Everybody's good at it, gay people are better at it than straight people, and, again, you don't need much time. You just need about a second. You see somebody for about a second, you could make a guess. You're far from always right. In fact, you're just a bit better than chance but you are better than chance at telling sexual orientation. So, these two facts taken together, thin slices and the power of first impressions, means that just by a brief exposure to somebody it shapes so much of how you're going to think about them in the future.

Now, we can look at this from the other direction. We're talking about the perceptions of other people, how we perceive other people, but social psychologists are also interested in the question of what happens to other people as a result of being perceived in a certain way. So, one question is, "What would cause me to perceive somebody as intelligent or stupid, gay or straight, anxious or level-headed?" A second question is, "What are the effects of being judged that way?" And psychologists have coined a term, talk about self-fulfilling prophesies, and the claim here more specifically is what's known as "the Pygmalion effect." And the Pygmalion effect is if I believe you have a certain characteristic this might cause you to behave as if you have that characteristic.

The name comes from the play by George Bernard Shaw Pygmalion, and the quote here is "The difference between a lady and a flower girl is not how she behaves but how she's treated. I shall always be a flower girl to Professor Higgins because he always treats me as a flower girl and always will," made into a better known movie, My Fair Lady. But I think that the same theme is better exemplified in a far better movie, La Femme Nikita, where a cold-blooded killer is treated with respect and affection and then she becomes a much more warm and accessible person, and then she kills a lot of people but that — [laughter] but still it illustrates the point.

And this point has tons of empirical validity. The classic experiment was by Rosenthal and Jackson where they told teachers that some of their kids were really smart and other kids were less — were not really smart, they weren't expected to show a huge jump or spurt in their IQ, and this was of course trickery. The children were chosen at random but the children who were described as showing — as expected to show a jump in IQ, in fact, did show a jump in their IQ scores and this isn't magic. It's basically — if I am told that you're a genius and your genius is about to be in full-flower throughout this class and it's a small class as these classes were, I'll focus more on you, I'll give you more of my attention. If I'm told "not so much for you," you'll suffer relative to him. And so the Pygmalion effect shows how our expectations can really matter.

Chapter 3. Positive Uses and Negative Effects of Stereotypes [00:11:15]

This brings us to the final — the issue of expectations and how we judge people is a story that could be told about individuals but it's also a story that could be told about groups. And this is where I want to end this section on social psychology by talking about groups. A lot of social psychology is concerned with the question of how we think about human groups and we've already discussed this in the lecture on morality when we talked about the human dynamic pushing us to think in terms of "us" versus "them" as shown in the Robber's Cave study and also shown in the minimal group research by Tajfel

showing that from a motivational, emotional standpoint it's not difficult for us to think in terms of "my group" versus "your group." And this way of thinking has real consequences for our emotional life, our affective life, and how we choose to distribute resources. What I want to talk about here though is a different aspect of how we think about human groups. I want to talk a little bit about stereotypes.

Now, "stereotypes" in English often just is a bad word. To have a stereotype is to be — is to have something wrong with you. You might say it's not good to have stereotypes. Psychologists tend to use the term in a broader sense. We tend to use the term to refer to information we have about categories and intuitions we have about the typicality, our frequency of certain features of categories. And it turns out that collecting information about categories is essential to our survival. We see novel things all the time and if we were not capable of learning and making guesses, educated guesses, about these novel things we would not be able to survive. So, when you see this object over here you categorize it as a chair and you recognize that you could probably sit on it. This apple is probably edible, this dog probably barks and has a tail and eat me — eats me and doesn't speak English. These are all stereotypes about chairs and about apples and about dogs. It doesn't mean they're logically true. This could be a vegetarian dog, a poison apple, an explosive chair, but [laughter] they're typically true. And if you were suddenly stripped of your ability to make generalizations, you'd be at a loss. You wouldn't know what to eat, how to interact. So, some sort of ability to record information and make generalizations is absolutely essential to making it through life.

What's interesting though is we also categorize types of people. So, we have stereotypes in our heads about men and women, about children, adolescents or adults, whites, blacks, Asians and so on. Now, this is not essentially a bad thing for a couple of reasons. First, some of these stereotypes are positive. You might have positive stereotypes about certain groups. You might believe some groups are unusually creative or intelligent. You might have a particularly positive stereotype about your own group even if your own group is Yale students or your own group is people from France or your own group is people from such and so college. You might have positive stereotypes. More importantly, we collect stereotypes about groups of people through much the same way we collect stereotypes about categories like chairs and apples and dogs. And so they're pretty often accurate.

When there are studies which ask people who is more likely to be a lawyer, someone who's Jewish or someone who is Hispanic, who is likely to be taller, somebody from Japan or somebody from Sweden, people can answer these things. They have their stereotypes that guide their answers, and the answers are not arbitrary or random. Their answers are often correct and often possessing stereotypes lets us make reasonable and correct generalizations about the world.

That's the sort of good news about stereotypes but there's also bad news. One problem is that they're not always accurate and there's a couple of factors that could lead them away from accuracy. One is what we talked about before regarding first impressions, which is a confirmation bias. If you believe that homosexuals are effeminate, that gay men are effeminate, then this is going to shape how you see future gay men. If you see an effeminate gay man, you'll probably say, "Ah, more evidence for my theory." If you see a man who is not effeminate, you might ignore it or say maybe he's not really gay after all. If you believe black men are criminals, then when you see a black man who is a criminal you'll chalk it down as support but you'll pay less attention to evidence that white men are criminals and some black men are not criminals. You won't look at this as a scientist objectively scanning data.

Rather, you'll be biased in certain ways. You'll be biased to put extra weight on the cases that support your theory and diminish cases that refute it.

Furthermore, our data is not always reliable. So — oh, and this is actually an example of this at work. It turns out in the world of classical music there's a stereotype of women being simply less proficient than men: they play smaller than men, they don't have the same force and they have smaller techniques, they're more temperamental and so on. If you asked somebody who was a judge, the judge would say, "Look. This is just the way things are. I'm not being biased at all." The test of this then is to have blind auditions where people do their auditions behind a screen so you can't tell whether they're man or a woman, or for that matter, white or black or Asian or whatever. It turns out when you do that women get hired far more suggesting that the stereotype is A, incorrect and B, has a real negative and unfair effect on people getting hired.

A second problem is – what I was talking about immediately before this – is some of our data are misleading so we get a lot of the information about the world from the media. The media would include television and movies but would also include plays and books and stories. And to the extent these portray an unrealistic or unfair or biased perception of the world we could construct stereotypes that are faithful to the data we're getting but the data is not representative. And so people, for instance, object to the fact that when there's Italian Americans on TV they're often members of the Sopranos, a mobster family. Throughout history Jews have been upset at the portrayal of Shylock in "Merchant of Venice," not a very nice guy. And often in response people who want to foster more positive views will often try to — will often put in representatives from other groups in unusual ways to make that point. Anybody here ever see the television showBattlestar Galactica? Okay. Who's that? He's the star of "Battlestar Galactica." You don't know because you're too young. In the original "Battlestar" — [laughter] and I hate you. [laughter] In the original "Battlestar Galactica," this was the star. This was the main character known as "Starbuck," who got transformed into a woman in the more recent one, a sort of example of how portrayals are shifting in interesting ways.

There's also, of course, moral problems over stereotypes. So, it's fine to judge chairs and apples and dogs based on the stereotypes. It's even fine to judge breeds of dogs. If I told you that I decided to buy a greyhound instead of a pit bull because I wanted a dog of a gentle temperament, nobody would scream that I'm a dog racist [laughter] involving — and — but honestly, it's a stereotype. Greyhounds are supposed to be more passive and gentle than pit bulls. I think it's a true stereotype but it's a stereotype nonetheless. But we have no problems when it comes to things like breeds of dogs with stereotypes. We have serious problems judging people this way. So, for instance, it's a moral principle that some of us would hold to that even if stereotypes are correct it is still immoral to apply them in day to day life. The term for this would be "profiling."

Now, it gets complicated because there are some cases where we do allow stereotypes to play a role. When you all go and get driver's licenses or when you did get driver's licenses you have to pay higher auto insurance premiums than I do. I think this is perfectly fair because young people like you get into a lot more accidents with your reefer and your alcohol [laughter] and so it is — now, some of you are saying "that's a stereotype." And it is a stereotype but it's a statistically robust one and nobody lines up to protest this. It's an acceptable stereotype to make a generalization from. On the other hand, what if insurance companies determined that people from Asia got into more accidents than people from

Europe? Would people be equally comfortable charging people from Asia higher rates of insurance? Almost certainly not. So, the issues are complicated as to what sort of generalizations we're — are reasonable to make and what aren't.

There's also a second problem. Stereotypes have all sorts of effects. Now, some of them are obvious effects. If people — for instance, if people pull you over while you're driving because you're black, this could have a huge effect on how you feel welcome in this society on race relations and so on. But some of the effects are more subtle and more interesting and you might not expect this. And this is some work done by the psychologist Claude Steele and his colleagues at Stanford. And the issue is called "stereotype threat." Imagine you have a math test and this is the front of the math test. Claude Steele made an interesting discovery. Here is how to make black people do worse on this math test. It's very simple. The finding is that if your race or your group has a negative stereotype associated with it in any particular domain, being reminded of it serves as a stereotype threat and hence damages your performance in all sorts of domains. If the stereotype is "your group doesn't do good in this," if I remind you that you're a member of that group immediately before doing it, your performance will drop. Now, you know how to make women do worse on math tests too, like that, and this has a demonstrative effect.

So, stereotypes are complicated and morally fraught things. When people study stereotypes they often make certain distinctions between three levels of stereotypes and this is nicely summarized here. There's "public." If I asked you: One of the people running for the Democratic nominee for president is black, another one is female — if I asked people to raise their hands for who thinks that because being black or because being female they should be automatically disqualified for being president, few of you would raise your hands. Those are your public presentations of stereotypes. Even if I was to ask you on a sheet of paper, you might deny it because you might be afraid that it's not anonymous. Then there is private. Private is what you really think but you don't tell people. Some of you — some of the population of the United States are not going to vote for somebody because he or she is black but they won't tell you but they know it to be true. That's common sense. What's more interesting is below even that there may be unconscious associations that work that people don't actually know about but affects their thoughts about race, gender and other social groups.

So, here are some data about what people publicly say. This is the proportion of people who say they will vote — they would vote for an African American president. What's interesting is when — around when I was born the answer in the United States was about half. Now, it is as close to one hundred percent as you can get it. Here is another one. This is also public stereotypes of blacks, proportion of white respondents endorsing each trait. Now, it's infinitesimal. These numbers are so low they could be dismissed as people filling in the wrong things or making jokes or just being confused compared to [laughter] stunningly superstitious high rankings. And so, there's been a profound change in public presentations, public views, on race but what about implicit views? This gets more complicated.

Here is a simple study. This is the sort of study that you might do here at Yale. What you might do, for instance, is be sitting at a computer screen and you'll be given incomplete words to fill out like "hos-" and you have to fill out this word. What you don't know is that pictures of black faces or pictures of white faces are being flashed on the screen but they're being flashed on the screen subliminally so fast you don't even know you're seeing them. Still this has an effect. When you see black faces subjects are

more likely to fill this with words like "hostile" while whites more likely to fill it with words like "hospital."

Chapter 4. Implicit Attitudes [00:27:19]

I will now welcome you to participate in an experiment on implicit attitudes. This was developed by Mahzarin Banaji who used to be at Yale and now is in an inferior university in Boston [laughter] and it's called implicit attitudes test and it's the biggest psychology experiment ever done in terms of people. I don't know. A million people have participated in this and you could just go online, implicit.harvard.edu and then you could do it yourself. But we'll do it now as a group. If you did it in the lab or on your computer screen, you would do it by pushing buttons. We'll do it by speaking. And it's very simple. You're going to see things over here and they're either going to be words or they're going to be pictures. If it's an African American or a bad word, a negative word, I want you to shout out "right," this side, "right." If it's a white American or a good word, I want you to shout out "left." People ready. Try to do it as fast as possible without making any mistakes. [audience response] [laughter] Because of the very loud wrong person we're going to try that again. [laughter] Are you ready? [audience response]

Good. That is "congruent," congruent according to a theory that says that people, both African Americans and white Americans, have biases to favor white Americans over African Americans. How do we know by this? Well, we compare it. That's "congruent." Now, it's different. If it's a white American or a bad word, say "right." If it's an African American or a good word, say "left." [audience response] Okay. For all I could tell, people did equally well but this experiment has been done tens of thousands of times and you could do it yourself on a computer screen. And this is one way of doing it but they'll alternate and they'll give you different ones to shift around and everything. And it turns out that this version people are slower at than the other version suggesting that their associations run one way and not the other. And this work has been extended for all sorts of ways looking at for example at gender, looking at the connection between women and English and men and math, looking at age, attitudes towards people who are obese versus people who are thin, attitudes towards people who are straight versus people who are gay, and you could go online and do these studies and it'll give you some feeling for the sort of implicit attitudes that we have within us.

Well, a legitimate question is "Who cares?" I mean, if you do the — If you look at the results for the study, it turns out that there is an association as bias to view white Americans as positive and African Americans as negative but it shows up in half a second difference. Who cares? Well, there's two answers to this. One answer is there are times in your life where half a second can matter a lot. So, studies with police officers using reaction time in split-second choices on who to shoot find that your stereotypical attitudes play a huge role in who you're likely to shoot when they're holding an object in their hand that's unclear. Also, more generally, it could be that these implicit attitudes play a role in judgment calls. In cases where you have a hard decision to make, you know you're not racist. You have no explicit racist attitudes, honestly you really don't, but the argument is that these stereotypes can affect your behavior in all sorts of subtle ways.

Here is one example. What they do is they do an experiment where somebody is in trouble. You hear a scream from outside either from a black person or a white person. In one condition you're the only

person around. In another condition there's other people around you. Now, we know from the work in the Bystander effect that in general which one are we more likely to help in, when we're the only person or multiple? [audience response] "Only," exactly. And in fact, when it's the only person just about everybody helps regardless of the color of the person in trouble but when you're with other people there's a big difference. Now, again this isn't — these things are not done with members of the KKK. They're done with the standard university undergraduates like you and these — and if you were in this group you wouldn't say, "Oh. I didn't want to help because the person's black." Rather, what you would say is, "Well, I didn't think it was worth helping. There were other people around. Someone else would help," but we know by looking at it that this difference makes a real difference.

A final study, and this was done by my colleague who just got hired here, Jack Dovidio, who's done some wonderful work, looked at how people judge to hire somebody based on their recommendations. This is a little bit of a confusing thing. The green bars are the African Americans. The blue bars are the white Americans. And in some cases these people have strong recommendations. When they have strong recommendations in 1989 you're willing to hire everybody, and this [the level of hiring for African versus white Americans, depicted on a slide] is not a difference at all. But when their recommendations are so-so, when it's a judgment call, the subjects are significantly more likely to hire the white American than the African American. Nineteen eighty-nine was a long time ago but the same results showed up in 1999. The same results also showed up about a year and a half ago. And again, this doesn't show that people are explicit terrible racists. It does show that people possess these stereotypes that make a difference in their real-world behavior.

A way to put this all together is in Trish Devine's automaticity theory, which goes something like this. The idea is that everybody holds stereotypes. These are automatically activated when we come into contact with individuals. In order to not act in a stereotyped fashion, we have to consciously push them down, we have to consciously override them, and that's possible, but it takes work. It takes work both at the individual level and it takes work at the group level.

Chapter 5. Question and Answer on Stereotypes [00:34:47]

Any questions or thoughts about this? Yes.

Student: [inaudible]

Professor Paul Bloom: It's a good question. These results are surprising and disturbing and I said there is work to be done at an individual and group level and this young man challenged me to be more explicit about that. Here is one case. We have job searches and sometimes job — senior job searches involve the faculty sitting in — around in a room and tossing out names and we use these names as a basis for further discussion. "Hey, I wonder — What about that person? That person does great work." What we do now in the psychology department is we make a special point of trying to get names from disadvantaged groups, not with an eye towards an affirmative action policy but rather because there's a lot of evidence suggesting that people who are just as qualified don't come to mind unless you make some sort of procedure to do it.

Here's another example. A lot of journals do blind reviewing now because of the evidence I talked about before regarding sexual stereotypes, that whether it has a male name or a female name makes a

difference. So those are not — those are group level in that they're not saying, "You get rid of your prejudices by trying harder." It's rather, "Let's set up a system so that your prejudices can't work," the blind auditions, for instance, being a beautiful example of that. At an individual level, your question's a harder one and I'm not exactly sure what we could do but I think what we should do is be conscious of these things and know that it's not enough to say, "Well, I'm explicitly not racist so I'll give everybody a fair shake," but to recognize we might have biases and to work hard to overcome them, not by overcorrecting in some random way but trying to — if you're interested for instance in qualifications setting up some system that these qualifications can be observed absent knowledge of race or sex. Yeah, in back.

Student: [inaudible]

Professor Paul Bloom: Yes. The question's a good one. How much of this is due to stereotypes and how much of this is due to an own-group bias? So, the fact that — so, the experiments as I have described them are to some extent ambiguous. The fact that white Americans favor white Americans over black Americans might be because of stereotypes but they also might be of in-group favoritism. I think the answer is that both play a role but some of the effects are due to stereotypes above and beyond in-group favoritism. And one reason why we know that is in studies like the IAT, the Implicit Attitudes Test. African Americans show much the same effect as white Americans. So, African Americans also are biased against African Americans and in favor of white Americans, showing it doesn't reduce to group favoritism though that probably plays a big role.

Chapter 6. The Minor Mystery of Sleep [00:38:09]

Okay. I'm going to shift and spend the rest of this class on a couple of mysteries. Here's a summary. The first one is a minor mystery. I think we have some progress in explaining it. The second one is a total mystery. The first one's sleep. Sleep is a motivation. It is a motivation like food or drink. It is a form of torture. I won't get into definitions of what torture is but it would cause somebody tremendous pain and anguish to keep them from sleeping. When you're really tired sleep is what you want to do like when you're really hungry you want to eat. How many people here on average — from the beginning of the semester until now get on average more than eight hours of sleep a night? That's good. Good.

There is a sort of school of sleep macho. A sleep macho used to be "I only get forty-two minutes of sleep a night." Now sleep macho is "I sleep eleven hours." [laughter] Who gets on average more than seven? On average more than six? Who here has been making it since the beginning of the semester on under six hours a night? Okay. Anybody of you that has been getting under five hours a night? Okay. There is big individual differences in how much sleep people need and sleep itself is what we spend a lot of our life on but it's very hard to study and we didn't used to know much of it because you can't ask people what's happening during it because they're sleeping so you need sort of clever methods. One such clever method is an EEG. You bring somebody in the sleep lab, you put electrodes on their scalp and you see what these — what sort of electrical activities you get in the brain. Right now many of you are showing irregular beta waves suggesting intense comprehension and great intellectual focus. [laughter] Some of you are awake but non attentive and your brain's giving you these large, regular alpha waves. Some of you are sound asleep, deep in delta. [laughter]

When you sleep you get the following stages. You start off with a transition period when you're falling asleep. We call that stage I. Then you get successively deeper, II through IV, slow, irregular, high amplitude delta waves, and then once you reach stage IV you start going up again, up through stage III and II. Then REM sleep emerges, rapid eye movement sleep. REM sleep is neat because your brain looks like it's wide awake but – and I'm going to talk a little bit more in detail about this later – you're relaxed, your rapid eye movements occur; that's where your — the name comes from and dreams occur and then on a good night's sleep you've got four to five sleep cycles and it looks like this. You start off and you go down, down, down, and then you come up, then you get your first REM cycle, again, again, again, again, again, and then you wake up.

So, the general takeaway message here is that there's two types of sleep. There is slow-wave sleep or "quiet sleep." Your eyes drift separately and slowly you're hard to wake up. Then there is REM sleep. Your brain is active as if you were awake, your EEGs are similar to waking, paralyzed except for the eyes, oh, men get erections and you have dreams. One sleep researcher joked that these two are connected because what happens is dreams fly around through the ether and then their erections serve as antennas so you pick them up. [laughter] Don't write that down. [laughter]

Why do we sleep and why don't we — why aren't we always awake? There's a couple of answers to this. Probably the best answer is our body is worn out during the day and sleep is necessary to put it back into shape. So, when you sleep growth producing hormone goes through the body, your brain and other organs get restored, there are — you need less food, your immune system seems to be hard at work, and in fact, one answer — people always wonder what happens if you'd stop sleeping and the answer is not one discovered in the laboratory but it's been recorded in different cases. If you stop sleeping you die. You don't die in some dramatic way. You get sick and then you die, suggesting that sleeping is good to keep you healthy. And an analogy I like to think about is that your — when you bring your car in to repair it — for repairs, when you're repairing your car, the first thing you do is you stomp it and shut off the engine. You make it stable and at rest so you can then work on repair. A related view is that sleep emerged to preserve energy and keep you out of trouble at night. And these views are of course not incompatible. This explains why we sleep at all and this probably explains why we sleep at night.

There are sleep disorders. I won't go through them. I just list a few of them here. They're sort of bad things that could happen to you when you're asleep. One particularly trendy sleep disorder that's been discovered recently are side effects of sleeping pills such as Ambien. So, one of the bizarre side effects is some people with Ambien while sleeping go downstairs, open up their refrigerator and eat huge amounts of food. [laughter] A more recent side effect of Ambien that's quite serious is some people become — have the compulsion while they're sleeping to go driving, which is not good. [laughter]

Chapter 7. The Greater Mystery of Dreams [00:44:49]

What we're really interested in though when it comes to sleep is dreams. So, remember Hamlet. People — you've all studied Hamlet, "to be or not to be." That's all I know but basically [laughter] he was deciding — he — I wrote this down though. He was deciding whether or not to kill himself and so he made up two lists. The lists of reasons to kill himself were "the slings and arrows of outrageous fortune," "a sea of troubles," and "the heartache in the thousand natural shocks that flesh is heir to."

He only had one argument not to kill himself: Nightmares. "To sleep, perchance to dream, for in that sleep of death what dreams may come when we have shuffled off this mortal coil might give us pause." He was worried about nightmares.

Now, dreams have always fascinated people and philosophers. If you take a philosophy course, there'll be a couple of questions about dreams that you will wonder about. One is "Are you dreaming right now?" So, Rene Descartes famously wondered whether the real world doesn't exist. Maybe right now it's just a dream. See also "The Matrix." And can you be immoral in a dream? We do all sorts of things in dreams that are bad. Are these sins? And the great theologians and philosophers like Augustine wondered about this, can you sin in a dream? I will answer both questions. You are not dreaming right now, unless you're asleep but if you're — and you can't usually be immoral in a dream. The exceptions are lucid dreams where you choose what to dream about, and then possibly you could be immoral if you encourage bad habits of thought.

What do we know about dreams? Well, first there's a distinction, a distinction between real dreams versus sleep thought. So, real dreams are you're in a submarine wrestling a chicken while your grandmother looks disapprovingly on. Those are real dreams. "Sleep thought" is the sort of thing you get typically before you fall asleep and it's like "did I take the garbage out, where is the garbage, did I take the garbage out?" And so it's really just this kind of rumination and you could do this while you're sleeping. So, if you're woken up during slow-wave sleep you're going to be thinking "Did I take the — yes, while I was dreaming I was thinking about the garbage," but if you're woken up during REM, "But a monkey was eating my grandmother," and that sort of thing. [laughter] So, there is a distinction.

What do people dream about? Well, we know some facts about dreams. Everybody dreams. Not everybody remembers their dreams. If you want to remember your dream by the way keep your dreams — keep a dream diary, very useful, but everybody dreams three to four times a night. That depends on how much sleep you get but dreams leave fragile memories. They fade quickly. This is why a dream diary or writing up — writing your dreams as soon as you wake up turns out to be useful.

What do people dream about? Well, basically, the way to find that out is you ask people and people have — psychologists and sociologists have collected dream reports. If you go to dreambank.net, a guy named Hill collected 50,000 dream reports; you could make some generalizations. Most dreams are bad. They're not bad, bad, they're not nightmares, but most dreams are from a scale of one to ten with five and a half in the middle they're on the negative side. They report misfortunes. People in tribal societies have dreams with more physical aggression than people in industrialized societies. Men have more aggressive dreams than women. Americans have more aggressive dreams than Europeans. Yeah. [laughter]

What do people want to dream about? Well, psychologist — the findings were totally not going to surprise you. Women want to dream about romance and adventure. Men want to dream about sex with strangers. [laughter] Turns out that once you get past the hormonal blast of adolescence about 10% of dreams have explicit sexual content.

What's the most common dream? Guesses. Falling. That's a good guess. Falling is among the top dreams but falling is not the winner. Somebody else. Flying? Also close. Public speaking? Good phobic dream but not it. Naked in public, not it. [laughter] Excellent. Being chased. Evolutionary

psychologists and cognitive neuroscientists have puzzled over this one. It seems to be sort of a primal dream but the dream about being chased is a biggie. Has anybody here ever dreamt of being chased that you could remember? Yeah. Being chased is the big one. Naked in public? [laughter] Grandmother killed by a monkey? [laughter] No.

So, what are dreams for? There are Freudian theories that dreams are disguised with fulfillment and other things. There is not much support for these. One theory which is popular is that dreams are a side effect of memory consolidation so your body, sort of below the neck, rebuilds itself while you're sleeping but also what happens while you're sleeping is your memories get played over and over again to consolidate them into different parts of the brain. Almost — the best analogy here is backing up a computer. Your brain backs itself up. In the course of backing itself up, there are sort of random events flash to consciousness and get put together in a coherent story. From this perspective, dreams serve no adaptive function at all but rather they're epiphenomena. They are the byproducts of another system.

Chapter 8. The True Mystery of Laughter [51:31]

Final topic is laughter and this is a true mystery. I got interested in this — well, I got interested in this as a developmental psychologist watching my own children laugh and trying to figure out what it is that made them laugh. This is my son, Zachary, pretty young [clip playing] but [laughter] this is not the youngest record of laughing. I found this on the web. [clip playing of an infant laughing] [laughter] It's a huge puzzle. From an evolutionary point of view, people's pursuit of sex and food and drink and sleep are not these huge mysteries, our abilities to understand language and make sense of the visual world and cohere in groups, but the fact that we make this weird noise at bizarre circumstances is a huge puzzle and typical — typically psychologists have failed to explain this puzzle.

So, here's something; here's a first guess I read in a neuroscience textbook. "We laugh when there is incongruity between what we expect and what we actually — what actually happens unless the outcome is frightening." [laughter] Now, this is a fair effort. It is about as wrong as anything can be wrong in any possible way. [laughter] The first thing is it doesn't really explain "why?" So, why should a non-frightening incongruity cause people to make a distinctive loud noise consisted of staccato segments of one fifteenth of a second each separated by a fifth of a second? It doesn't explain why we make that loud noise when dealing with incongruity. It's not the case that incongruity causes laughter. If there was a bowl of fruit up there as you walked in, it'd be incongruous but people wouldn't shriek with laughter and point to the fruit. [laughter] It's also — a lot of laughter isn't caused by incongruity so a lot of times when we laugh there's nothing incongruous in any deep sense about it. So, laughter is kind of a puzzle. We don't know why we laugh. We don't know what makes us laugh.

Over the last five, ten years there's been some work done on this. A lot of the work is done by Robert Provine who summarizes this in his excellent book Laughter. And Provine decided to do something that nobody in the history of the world has done. He decided to send himself and his students to different places including shopping malls and observe when people laughed and write down what made them laugh. And this is the first descriptive step to developing a theory of laughter. And his big finding is that you could separate people — you could separate the question of laughter from the question of a joke or the question of humor. Most of what people — what made people laugh wasn't in any sense a joke or humorous.

So, he had over a thousand laughter-initiated situations and here are some typical comments that initiated laughter, sometimes uproarious laughter, on the parts of people. "I'll see you guys later." "Look. It's Andre." [laughter] "Are you sure?" "I know." "How are you?" "I try to lead a normal life." It's sort of funny. [laughter] It was anyway in a context where it wasn't particularly amusing. "It wasn't you." "We can handle this." Only ten percent of the comments were — could be re-coded later on as people — as actually humorous in any sense. And these included "Poor boy looks just like his father," [laughter] "You smell like you've had a good workout," [laughter] "Did you find that in your nose?", [laughter] a reference to dormitory food, [laughter], "He has a job pulling back skin in the operating room." [laughter]

So, Provine suggests we separate the question of laughter and jokes and then ask what do we know about laughter? Well, here are some basic facts that — remember we're trying to think like evolutionary biologists here. Humans have this universal trait. There's no society without laughter. It's early emerging as we saw in those clips. So, what's it for? What does it do? If it's an accident, what's it an accident from? What properties does it have? And there are some properties that are important to know. It's social and communicative. Why do we know that? Because it's loud. Laughter is not like hunger. Hunger can be silent. Hunger is not essentially a message to other people but when humans – when it involves a loud noise – the reason why we've evolved loud noises is to communicate with other people.

Laughter could be viewed as a form of involuntary noisemaking and it's contagious. That's another interesting thing about it. One of the great discoveries in television and movies was the invention of the laugh track. The laugh track makes things a lot funnier because people laugh a lot more when they're in groups. The rim shot after a comic remark is an attempt to simulate — sort of a pre-technological attempt to simulate the sound of laughter. Children are particularly influenced by the contagiousness of laughter, but just in general, if you hear somebody laughing, like the kids you saw before, it's the sort of thing that could easily make you laugh.

Other primates do it to some extent and then it's interesting when they do it. Monkeys laugh when they attack. When monkeys get together to kill and eat somebody they make a kind of a laughing noise and chimpanzees laugh when they tickle each other.

And then you'll see — And what's tickling? Well, a fair definition of tickling is touching parts of the body in a mock attack. You sort of attack somebody and they're laughing but if it's — but if somebody's trying to mug you you're not "Ah, this is so funny." [laughter] You have to realize that this is a mock attack; you're simulating attack. So one theory is — is a signal of mock aggression and collective aggression. It's basically — it's like a sound of some sort of mob attack. It's mock aggression in the sense that when people laugh they're often teasing, kidding around. They're throwing out insults, maybe they're tickling each other, maybe they're making fun of each other, maybe they're making fun of themselves, and there's some aggression to it. Most laughter is inspired by some degree of aggression, but it's attenuated and it's not real. And the laughter is a signal this isn't real. And collective aggression — a lot of mob assaults, executions, lynchings, rapes are accompanied with the sound of laughter. When you're in a group doing something terribly aggressive and you are not — you don't feel like you're at risk you might laugh and laughter is a signal to other members of your group as a signal of solidarity and what you're doing together.

A different way of putting it — and none of this is going to come in with a sharp, decisive theory. These are sort of flailing away at different ideas but another way of doing — an older way is proposed by Plato, who viewed laughter as a form of bonding against a common enemy. It's a sound of group cohesion against the common enemy. Comedians have not missed the aggressive nature of laughter. Dave Barry writes:

The most important humor truth of all is that to really see the humor in a situation you have to have perspective. Perspective is derived from two ancient Greek words, ‘pers' meaning something bad that happens to someone else and ‘ective' meaning ideally somebody like Donald Trump.

[laughter] And the idea is that it's something bad happening but not to you or to somebody you love. Mel Brooks puts it in somewhat sharper terms, distinguishing between comedy and tragedy: "Tragedy is when I cut my finger. Comedy is when you fall in an open sewer and die." [laughter]

To sum up, ingredients of humor is there has to be a target who experiences some harm. It could be an enemy. It could be a friend. It could be yourself. The harm shouldn't be so serious that it elicits strong negative emotions like fear, grief or pity. So, if a stranger slips on a banana peel and lands on his butt, I might laugh because I'm not overcome by compassion. If he splits his head open and there's blood everywhere, I'm less likely to laugh unless I don't like him or something. [laughter] And this is why the humor — why the damage is often a certain sort of damage involving things like embarrassment, sex, scatology, a banana peel, pie in the face, your pants fall down or something, where there's no real harm. So, empathy, caring, sympathy don't kick in, but instead there's the aggression unleashed at somebody and there has to be some level of surprise.

This is what — this is where the humor for that baby comes in, which is you couldn't predict when the sound was coming and some aspect of that was what made it so funny then, what elicited the laughter. This is — what elicited the laughter for Zachary was watching the classic film Winnie the Pooh and the Blustery Day where Rabbit climbed on a high bunch of shelves and it all came crashing down on him. But he didn't — he wasn't dead. He was just kind of stunned. [laughter]

Final question for anybody interested in laughter is, "Why can't we tickle ourselves?" Now, I've talked about laughter on two separate occasions and on each of the occasions when I talked in front of a large group like this somebody came up to me afterwards and said, "But I can." [laughter] And [laughter] I don't — and they seemed sincere but if people say, "I can tickle myself," how many people here will own up to being able to tickle themselves? One? [laughter] It's a fascinating question why some people can. The general story of why we can't in general is because there's no surprise, there's no mock aggression, and also there may be a general deadening of self-stimulation.

I'm going to end with the final reading response. Think of an interesting testable idea about either dreams or laughter. You could go into as much detail as you want. If you want your thing could be one sentence long but it could be longer. I will ask the teaching fellows to each send me their one or two most interesting remarks. I will then judge and then the winner will win some small prize, either of a literary or a food nature. Okay. I'll see you all on Wednesday.